UNIVERSIDADE LUSÓFONA

GeekLab:
Build a Home Lab (Like a Pro)

João Caldeira

Assistant Professor

Email. joao.caldeira@ulusofona.pt

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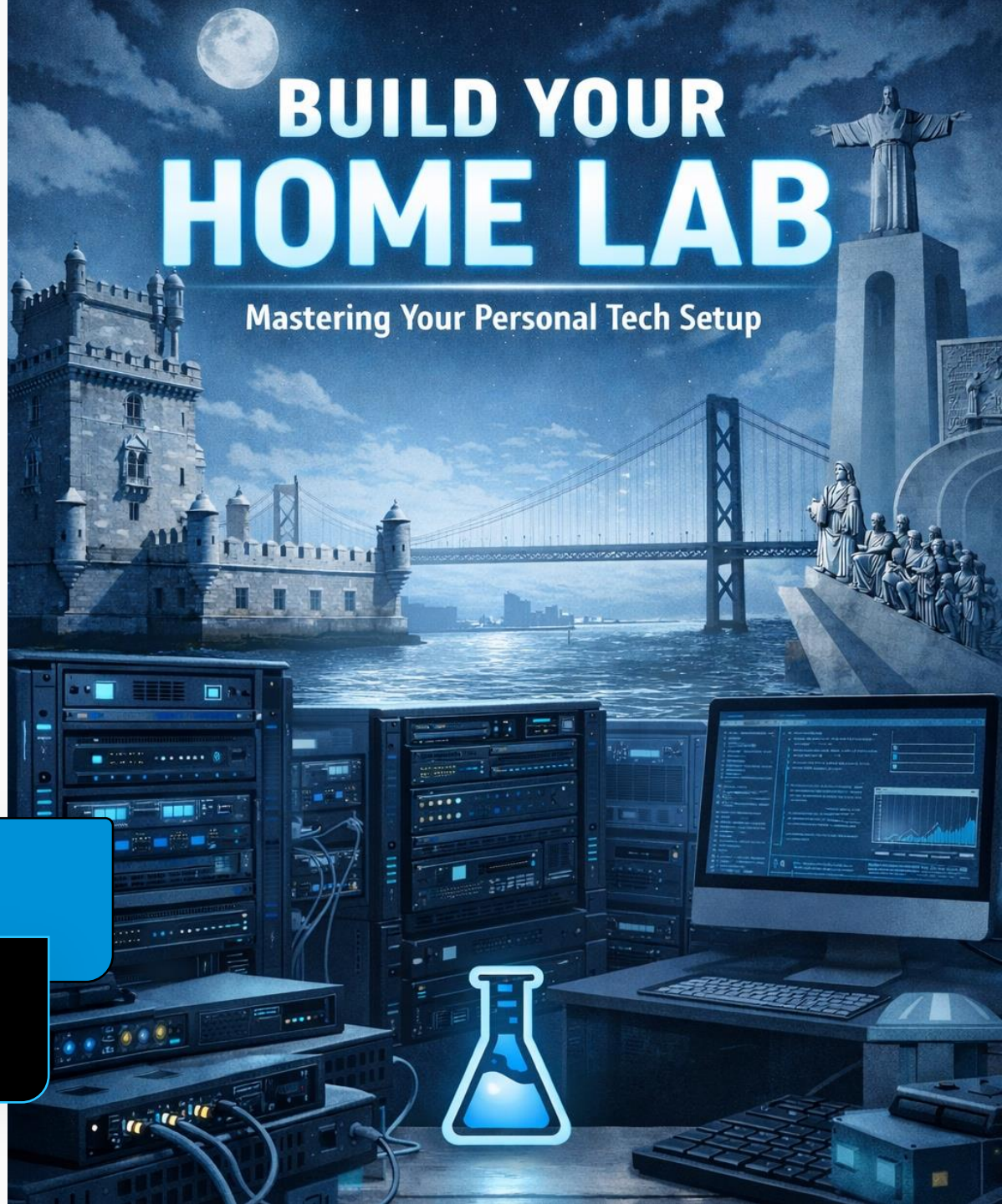

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Requirements

1. **Having a Physical Machine**
   a) **Personal Laptop / Desktop / Tower**
   b) **Dedicated Server**
   c) **Dedicated RaspberryPI**

2. **Docker installed on that machine**
3. **Docker Container Management Tool (as first container)**
4. **Deploy a Proxy Server**
5. **Deploy a few Application Services**
6. **Deploy a Home Dashboard**
7. **Configure DNS** **(this should be the first thing to do, but in this workshop, we will dot it at the end)**

Github Repository: https://github.com/jcppc/tecweb26

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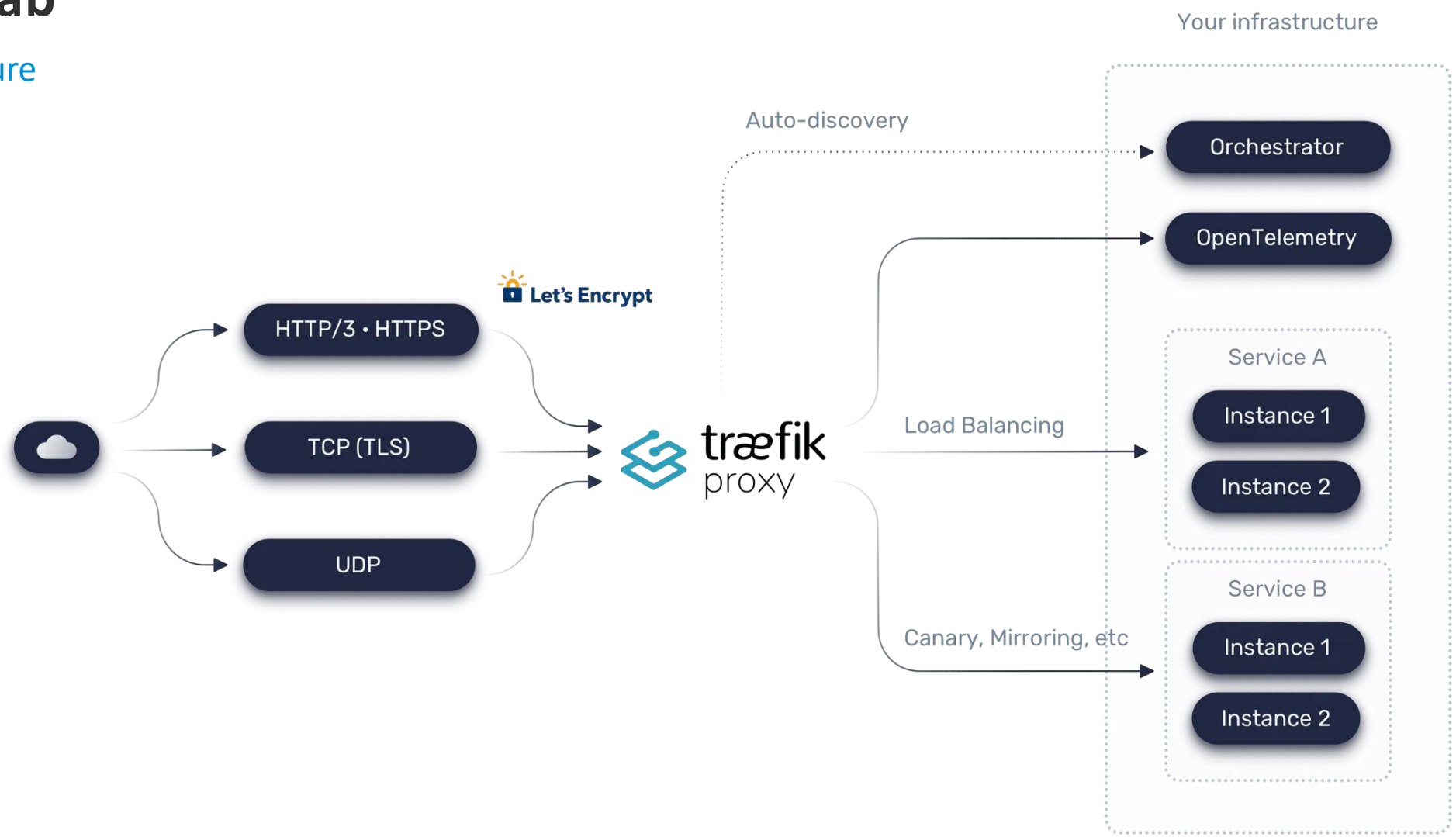
Architecture



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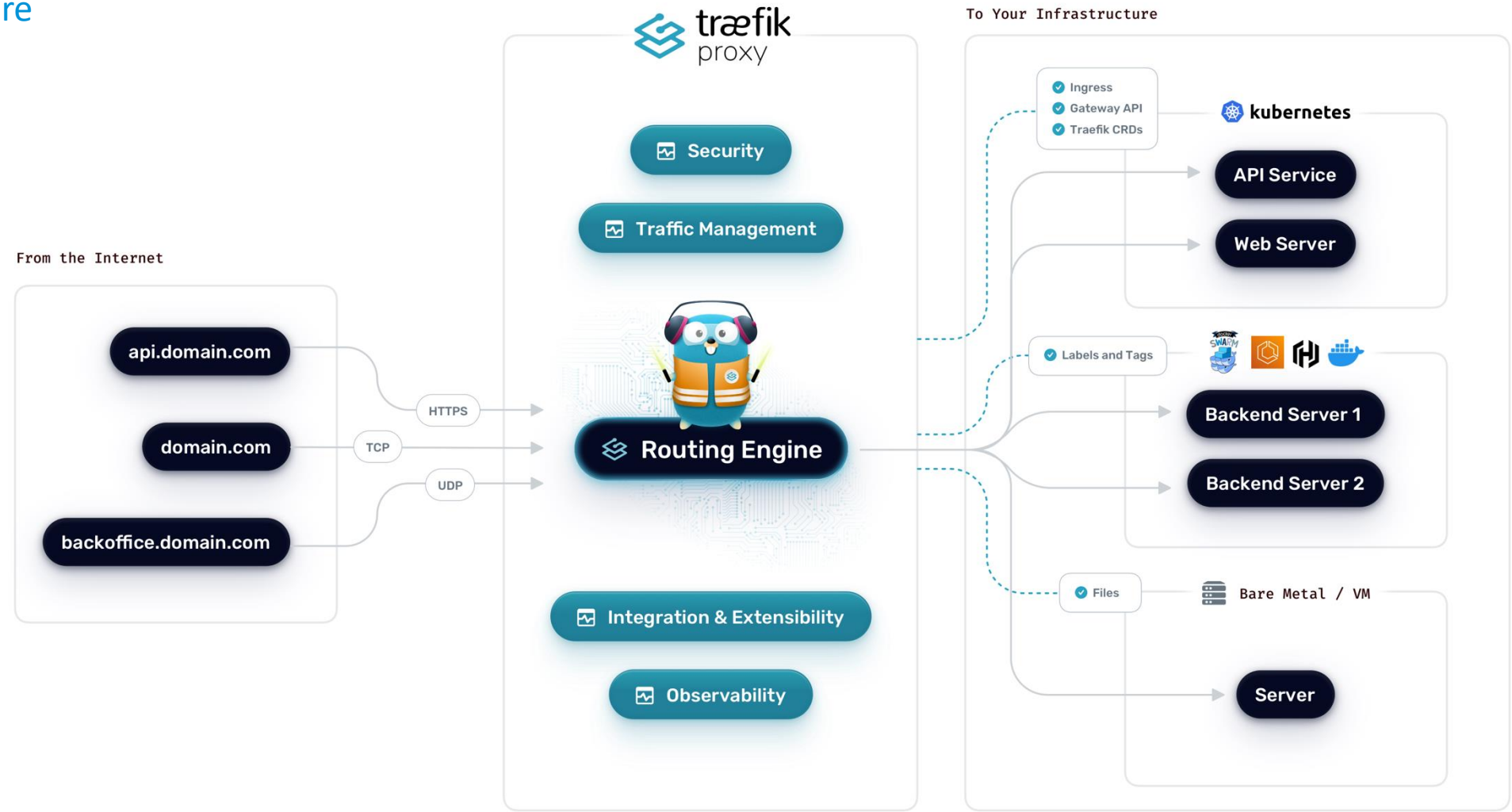
Architecture



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Requirements

1. **Personal Laptop / Desktop / Tower**

2. **Dedicated Server**

3. **Dedicated RaspberryPI**

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Requirements

1. **Personal Laptop / Desktop / Tower**
    a. Having an installed OS (Windows, Linux, macOS, etc)
    b. Configure static IP (dynamic IP also does the job)
    c. Install Docker Engine (eg. Desktop)
    d. Install Docker Compose plugin
    e. Create baseline directories

```
C:\lab
  ├── traefik/
  ├── pihole/
  ├── portainer/
  ├── dashboard/
  └── apps/
```

```
/home/lab/
  ├── traefik/
  ├── pihole/
  ├── portainer/
  ├── dashboard/
  └── apps/
```


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Requirements

2. **Dedicated Server**
    a. Install OS (Ubuntu Server / Debian / etc)
        a. https://www.youtube.com/watch?v=n7aEcfDNULc
    b. Configure static IP
    c. Install Docker Engine (for your OS)
    d. Install Docker Compose plugin
    e. Create baseline directories

```
/home/lab/
  ├── traefik/
  ├── pihole/
  ├── portainer/
  ├── dashboard/
  └── apps/
```

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Requirements

3. **Dedicated RaspberryPI**
    a. Install OS (Ubuntu Server / Debian / etc)
        a. https://www.raspberrypi.com/software/
    b. Configure static IP
    c. Install Docker Engine
        a. https://pimylifeup.com/raspberry-pi-docker/
        b. (https://www.youtube.com/watch?v=xc8Q7Hjyv2Q)
    d. Install Docker Compose plugin
    e. Create baseline directories

```
/home/lab/
├── traefik/
├── pihole/
├── portainer/
├── dashboard/
└── apps/
```


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Step 0 – Define a hostname for your laptop/server

- Add a **hostname** entry to your hosts file
  - **YOUR_HOST_IP_ADDRESS short_name FQDN -> this means Fully Qualified Doman Name**
    - 192.168.1.130 lab lab.tecweb26.com
    - **Test it with the following command line:**
      - ping lab
      - ping lab.tecweb26.com
  - **How to add host entries on Windows, Linux, macOS**
    - https://www.swhosting.com/en/comunidad/manual/modify-hosts-file-on-windows-mac-and-linux

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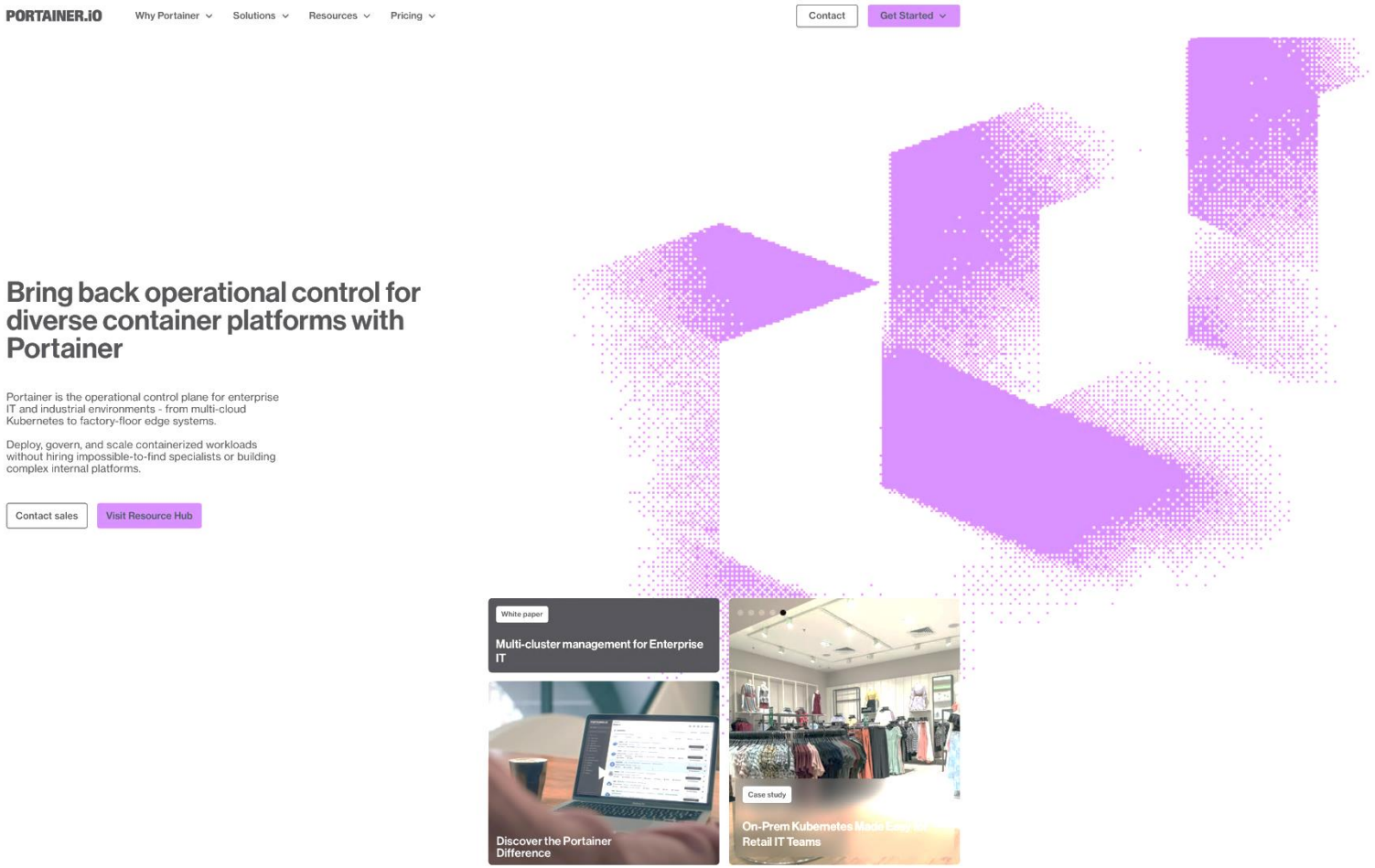
Step 1 – Install a Docker Container Management Tool

- **Portainer.io**
  - **Manage Docker containers using a web interface**



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Step 1 – Install a Docker Container Management Tool

- **Portainer.io**
  - **Install de Community Edition**
  - **On your system command line run:**
    - docker volume create portainer-data
    - docker run -d -p 8000:8000 -p 9443:9443 --name portainer --restart=always -v /var/run/docker.sock:/var/run/docker.sock -v portainer-data:/data portainer/portainer-ce:latest
    - Access web page on https://localhost:9443 (later we will open it on: https://lab.tecweb26.com:9443)
    - Create an admin **username and password**
  - **Extra info on:**
    - **https://arit-lusofona.kiion.com/portainer.html**

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Step 2 – Install a Reverse Proxy & Ingress

- **Traefik**
  - **On your system command line run:**
    - docker network create traefik
  - Access **Portainer** web page
  - Go to **Stacks**
  - Create a new stack
    - Name it: **traefik**
    - Copy code from Github **REPOSITORY** or **DONTPAD**
      - https://github.com/jcppc/tecweb26
      - https://dontpad.com/tecweb2026

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Step 3.1 – Install Core Application Services

- **Stirling** – A web application to manage pdf files (Edit, merge & cut pages, etc)
  - Access **Portainer**
  - Go to **Stacks**
  - Create a new stack
    - Name it: **pdf**
    - Copy code from Github **REPOSITORY** or **DONTPAD**
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Step 3.2 – Install Core Application Services

- **Langflow** – A web application to develop AI Agents
  - Access **Portainer**
  - Go to **Stacks**
  - Create a new stack
    - Name it: **agent**
    - Copy code from Github **REPOSITORY** or **DONTPAD**
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Step 3.3 – Install Core Application Services

- **DBeaver** – A web application to manage databases
  - Access **Portainer**
  - Go to **Stacks**
  - Create a new stack
    - Name it: **dbeaver**
    - Copy code from Github **REPOSITORY** or **DONTPAD**
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Step 4 – Install a Homelab Dashboard / UX Layer

- **Heimdall**
  - Access **Portainer**
  - Go to **Stacks**
  - Create a new stack
    - Name it: **heimdall**
    - Copy code from Github **REPOSITORY** or **DONTPAD**
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Step 5 – Install a DNS Foundation

- **Pi-Hole**
  - Access **Portainer**
  - Go to **Stacks**
  - Create a new stack
    - Name it: **pihole**
    - Copy code from Github **REPOSITORY** or **DONTPAD**
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Step 5 – Install a DNS Foundation

- **Pi-Hole**
  - Change DNS Configuration
    - **Windows**
      - https://www.youtube.com/watch?v=zXEqaM5IbYQ
    - **Linux**
      - https://www.youtube.com/watch?v=d_7vm6cKPjQ
    - **macOS**
      - https://www.youtube.com/watch?v=zRNbiN4u7E8&t=9s

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Step 5 – Install a DNS Foundation

- **Pi-Hole**
  - Add entries to hosts file
    - **Windows, Linux, macOS**
      - https://www.swhosting.com/en/comunidad/manual/modify-hosts-file-on-windows-mac-and-linux

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Assistant Professor

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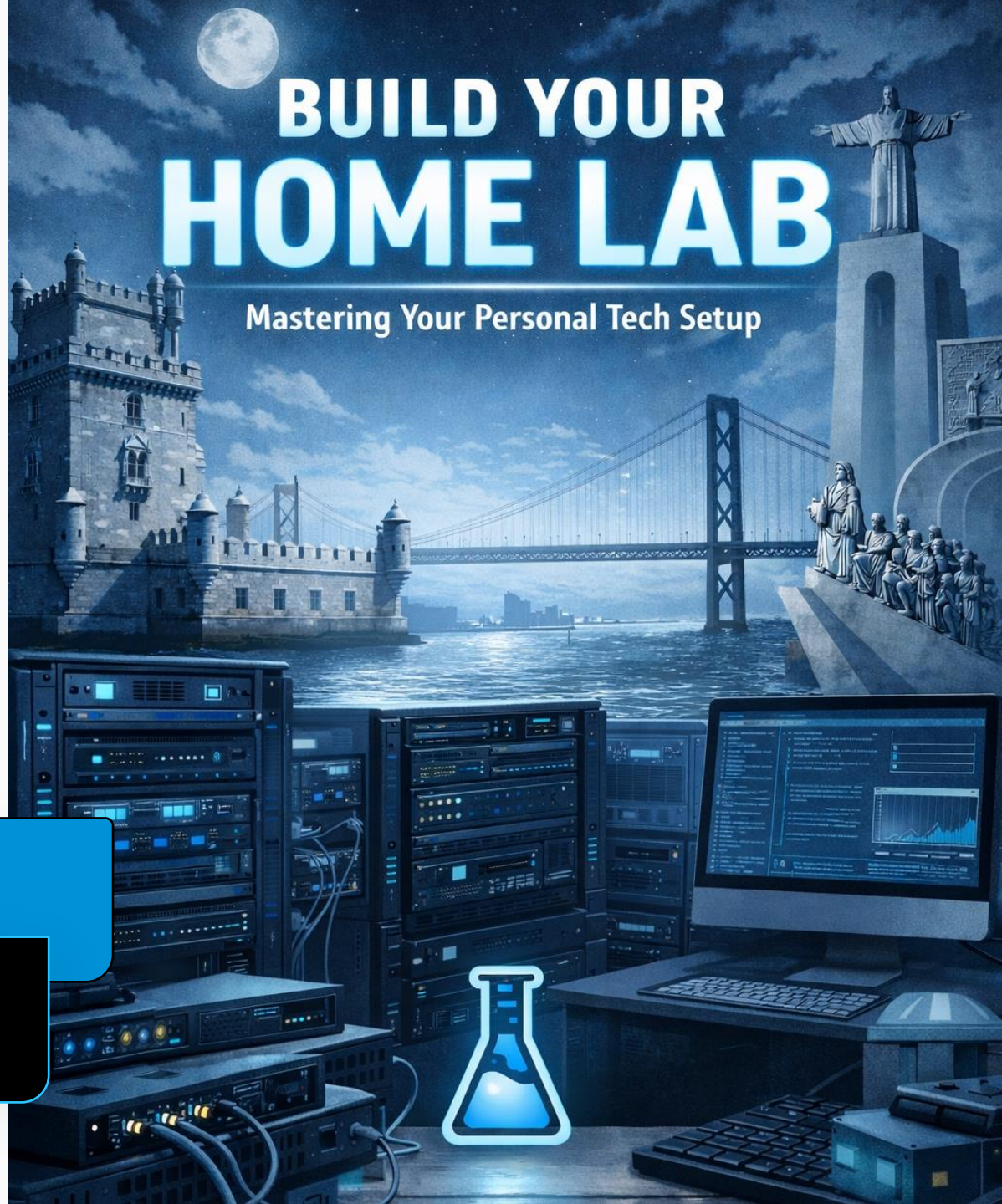
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